

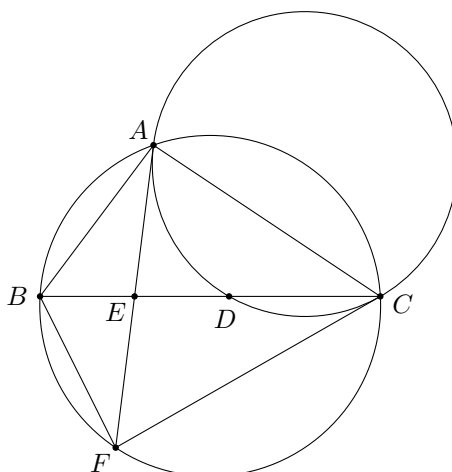


HK IMO Training 2023-2024

Problem Set 12 Suggested Solutions



Problem 45 (Kosovo MO 2022 Grade 11 Problem 4). Let D be a point on the side BC of $\triangle ABC$ such that the tangent at A to the circumcircle of $\triangle ACD$ bisects BD . Prove that $\sqrt{2}BC \geq AB + AC$.



Solution. Suppose the tangent at A to (ACD) intersects BC and (ABC) at E and F respectively. As

$$EA^2 = ED \times EC = EB \times EC = EA \times EF,$$

we know that $EA = EF$. By Ptolemy's theorem and similar triangles, we have

$$BC \times AF = AB \times CF + AC \times BF = AB \left(AB \times \frac{EF}{BE} \right) + AC \left(AC \times \frac{EF}{CE} \right).$$

As $AF = 2EF$, this yields

$$2BC = \frac{AB^2}{BE} + \frac{AC^2}{CE} \geq \frac{(AB + AC)^2}{BE + CE} = \frac{(AB + AC)^2}{BC}$$

by the Cauchy-Schwarz inequality. Thus, $2BC^2 \geq (AB + AC)^2$, which means $\sqrt{2}BC \geq AB + AC$. $\square$

Problem 46 (Bundeswettbewerb Mathematik 2016 Round 1 Problem 4). There are 33 students in a class. Each student writes down two numbers on the blackboard. The first number indicates the number of other students in the class who have the same surname as this student, and the second number indicates the number of other students in the class who have the same first name as this student. After all students have written down the numbers, it is found that each of the numbers $0, 1, 2, \dots, 10$ appears at least once on the blackboard. Prove that there are two students in the class who have the same surname and the same first name.

Solution. If a number k appears on the blackboard, there are $k + 1$ students having a particular surname or first name. All of them would write down the number k on the blackboard. Therefore, the number k must occur at least $k + 1$ times on the blackboard. Since there are $33 \times 2 = 66$ numbers in total and $1 + 2 + \dots + 11 = 66$, each of the numbers $k = 0, 1, 2, \dots, 10$ appears exactly $k + 1$ times on the blackboard, and each group of such numbers corresponds to a distinct surname or first name. Thus, there are exactly 11 different surnames and first names.

WLOG assume there are 11 students having the same surname (which corresponds to the number 10 on the blackboard). As there are at most 10 different first names (as one of the 11 names is used as a surname), two of these students must share the same first name by the pigeonhole principle. Therefore, the two students have the same name. $\square$

Problem 47 (Kosovo TST 2020 Problem 4). For each positive integer n , let $\varphi(n)$ be Euler's totient function, and let $\pi(n)$ be the number of prime numbers less than or equal to n . Prove that

$$\pi(m) - \pi(n) \leq \frac{(m-1)\varphi(n)}{n}$$

for all positive integers m and n . When does equality hold?

Solution. If $n = 1$, the inequality is the same as $\pi(m) \leq m - 1$. This obviously holds because the only possible prime numbers less than or equal to m are $2, 3, \dots, m$. Equality holds when all these integers are primes. As 4 is not a prime, this holds iff $m = 1, 2, 3$.

If $n > 1$ and $n \geq m$, we have $\pi(m) - \pi(n) \leq 0 \leq \frac{(m-1)\varphi(n)}{n}$. Equality holds when $\pi(m) = \pi(n)$ and $m = 1$ (since $\varphi(n) > 0$). This is impossible as $n > 1$.

If $m > n > 1$, then $\pi(m) - \pi(n)$ counts the number of primes p satisfying $n < p \leq m$. For any such prime p , we can write $p = kn + r$ where $k \geq 1$ and $0 \leq r < n$. Note that $r \neq 0$ since otherwise $p = kn$ cannot be a prime (note that $p > n$). Thus, we have $r = 1, 2, \dots, n-1$. Also, we have $(r, n) = (p, n) = 1$. For each r satisfying $1 \leq r \leq n-1$ and $(r, n) = 1$, the inequality $kn + r \leq m$ is the same as $k \leq \frac{m-r}{n}$. Thus, there are at most $\left\lfloor \frac{m-r}{n} \right\rfloor$ possible values of k . This gives at most $\left\lfloor \frac{m-r}{n} \right\rfloor$ possible values of p . Therefore, we have

$$\pi(m) - \pi(n) \leq \sum_{\substack{r=1 \\ (r,n)=1}}^{n-1} \left\lfloor \frac{m-r}{n} \right\rfloor \leq \sum_{\substack{r=1 \\ (r,n)=1}}^{n-1} \frac{m-r}{n} \leq \sum_{\substack{r=1 \\ (r,n)=1}}^{n-1} \frac{m-1}{n} = \frac{(m-1)\varphi(n)}{n}.$$

In order that the equality holds, we must have $r = 1$ and $n \mid m-1$. Note that $r = 1$ means the only integer from 1 to $n-1$ which is relatively prime to n is 1. As $(n-1, n) = 1$, we must have $n-1 = 1$, i.e. $n = 2$. Then m is odd. Also, all integers $2k+1 \leq m$ have to be primes. Since 9 is not a prime, this cannot hold if $m \geq 9$. Thus, it suffices to check $m = 3, 5, 7$. It is easy to check that all of $(m, n) = (3, 2), (5, 2), (7, 2)$ are equality cases.

In conclusion, equality holds when $(m, n) = (1, 1), (2, 1), (3, 1), (3, 2), (5, 2), (7, 2)$. □

Problem 48 (Serbia MO 2019 Problem 6). Two sequences (a_n) and (b_n) are defined as follows:

$$\begin{aligned} a_0 &= 0, & a_1 &= 1, & a_{n+1} &= \frac{2018}{n}a_n + a_{n-1}, \\ b_0 &= 0, & b_1 &= 1, & b_{n+1} &= \frac{2020}{n}b_n + b_{n-1} \end{aligned}$$

for all $n \geq 1$. Prove that $\frac{a_{1010}}{1010} = \frac{b_{1009}}{1009}$.

Solution. Let k be a positive integer. Define a sequence (x_n) by

$$x_0 = 0, \quad x_1 = 1, \quad x_{n+1} = \frac{2k}{n}x_n + x_{n-1}$$

for all $n \geq 1$. We claim that x_n is the coefficient of t^k in $(t+2)^{k-1}(t+1)^n$ for all $n \geq 0$.

When $n = 0$, the coefficient of t^k in $(t+2)^{k-1}$ is 0 = x_0 . When $n = 1$, the coefficient of t^k in $(t+2)^{k-1}(t+1)$ is 1 = x_1 . Assume the claim holds for the cases $n-1$ and n . We show that it also holds for the case $n+1$. Firstly, note that

$$x_n = \sum_{i=0}^{k-1} 2^i \binom{k-1}{i} \binom{n}{i+1}$$

as we can choose $k-1-i$ copies of t from $(t+2)^{k-1}$ and $i+1$ copies of t from $(t+1)^n$, where $i = 0, 1, \dots, k-1$. This implies

$$\begin{aligned} \frac{2k}{n}x_n &= \sum_{i=0}^{k-1} 2^{i+1} \frac{k}{n} \binom{k-1}{i} \binom{n}{i+1} \\ &= \sum_{i=0}^{k-1} 2^{i+1} \frac{k}{n} \cdot \frac{(k-1)!}{i!(k-1-i)!} \cdot \frac{n!}{(i+1)!(n-i-1)!} \\ &= \sum_{i=0}^{k-1} 2^{i+1} \cdot \frac{k!}{(i+1)!(k-1-i)!} \cdot \frac{(n-1)!}{i!(n-i-1)!} \\ &= \sum_{i=0}^{k-1} 2^{i+1} \binom{k}{k-1-i} \binom{n-1}{i} \\ &= \sum_{j=1}^k 2^j \binom{k}{k-j} \binom{n-1}{j-1} \end{aligned}$$

using the substitution $j = i + 1$. This is equal to the coefficient of t^k in $t(t+2)^k(t+1)^{n-1}$, since we can choose $k - j$ copies of t from $(t+2)^k$ and $j - 1$ copies of t from $(t+1)^{n-1}$. Recall that x_{n-1} is the coefficient of t^k in $(t+2)^{k-1}(t+1)^{n-1}$. Therefore, $x_{n+1} = \frac{2k}{n}x_n + x_{n-1}$ is the coefficient of t^k in

$$t(t+2)^k(t+1)^{n-1} + (t+2)^{k-1}(t+1)^{n-1} = (t+2)^{k-1}(t+1)^{n-1}[t(t+2) + 1] = (t+2)^{k-1}(t+1)^{n+1}.$$

This establishes the case $n + 1$. The claim holds by induction.

It follows from the claim that

$$\frac{a_{1010}}{1010} = \sum_{i=0}^{1008} 2^i \cdot \frac{1}{1010} \binom{1008}{i} \binom{1010}{i+1} = \sum_{i=0}^{1008} 2^i \cdot \frac{1}{1010} \cdot \frac{1008!}{i!(1008-i)!} \cdot \frac{1010!}{(i+1)!(1009-i)!}$$

and

$$\frac{b_{1009}}{1009} = \sum_{i=0}^{1009} 2^i \cdot \frac{1}{1009} \binom{1009}{i} \binom{1009}{i+1} = \sum_{i=0}^{1008} 2^i \cdot \frac{1}{1009} \cdot \frac{1009!}{i!(1009-i)!} \cdot \frac{1009!}{(i+1)!(1008-i)!}.$$

It is easy to see that the two summands are identical. □